

AI511/MM505 Linear Algebra with Applications

Lecture 4 – Inverses (cont.), Determinants

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September 9, 2025

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Schedule

Week	Lectures	Exercise sessions	Part
36	AI511/MM505	No exercise sessions	Theory
37	AI511/MM505	AI511/MM505	Theory
38	AI511	AI511	Applications
39	AI511/MM505	AI511/MM505	Theory
40	AI511/MM505	AI511/MM505	Theory
41	AI511/MM505	AI511/MM505	Theory
42	Autumn break		
43	AI511/MM505	AI511/MM505	Theory
44	AI511	AI511/MM505	Applications/theory

- Week 38 is a short introduction to linear algebra in Python.
- Week 44 is split: lectures cover applications, exercises cover theory.

Outline

Recap

Matrices and matrix operations (cont.)

- Inverse of a matrix (cont.)

- Systems of linear equations revisited

- Diagonal and triangular matrices

- Determinant

- Computing determinants

- What can determinants be used for?

Recap

Systems of linear equations and the inverse of a matrix

- A system of linear equations can be compactly written as

$$Ax = b, \tag{1}$$

where $A \in \mathbb{R}^{m \times n}$, $x \in \mathbb{R}^{n \times 1}$, and $b \in \mathbb{R}^{m \times 1}$.

- If $b = 0$, the system is called *homogeneous* and always has at least one solution.
- Can we somehow “divide” both sides of the equation by A so that we obtain the solution? What kind of a matrix does A have to be so that we could “divide” both sides of the equation?

Inverse of a matrix

- We say $A \in \mathbb{R}^{n \times n}$ is *invertible* if there exists a matrix, which we denote A^{-1} and call the *inverse* of A , such that

$$AA^{-1} = A^{-1}A = I_n.$$

- Note that the definition of the inverse of a matrix is only applicable to square matrices.
- If a matrix is invertible, its inverse is unique.
- Suppose that

$$A = \begin{bmatrix} 2 & -5 \\ -1 & 3 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}.$$

Which of these two matrices is invertible? Is there anything peculiar about one of these two matrices?

Characterising invertibility

Theorem (Theorem 2.2.4 of Johnston (2021))

Suppose that $A \in \mathbb{R}^{n \times n}$. The following are equivalent.

- (a) A is invertible.*
- (b) The linear system $Ax = 0$ has a unique solution.*
- (c) The reduced REF of A is I_n .*

This theorem does not tell us how to find the inverse of A .

Elementary matrices

Definition

A square matrix $E \in \mathbb{R}^{n \times n}$ is called an *elementary matrix* if it can be obtained from the identity matrix via a single elementary row operation.

Examples

$$\begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -3 & 0 & 1 \end{bmatrix}, \quad \text{and} \quad \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1/2 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

$R_1 \leftrightarrow R_2$ $R_3 - 3R_1$ $\frac{1}{2}R_2$

Matrices and matrix operations (cont.)

Matrices and matrix operations (cont.)

Inverse of a matrix (cont.)

Elementary row operations as matrix multiplication

Theorem (Theorem B.3.1 of Johnston (2021))

Suppose $A \in \mathbb{R}^{m \times n}$. If applying a single elementary row operation to A results in a matrix B , and applying that same elementary row operation to I_m results in a matrix E , then $EA = B$.

For example,

$$\begin{array}{ccc} \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} & \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix} & = & \begin{bmatrix} 4 & 5 & 6 \\ 1 & 2 & 3 \\ 7 & 8 & 9 \end{bmatrix} \\ =E & =A & & =EA \end{array}$$

Gauss-Jordan elimination as matrix multiplication

- Multiplying on the left by an elementary matrix is equivalent to performing a single elementary row operation.
- Multiplying on the left by several elementary matrices, one after another, is equivalent to performing a sequence of elementary row operations.
- This implies that the row operations in Gaussian or Gauss-Jordan elimination can be represented as matrix multiplication.

Finding the inverse of a matrix

- According to Theorem 2.2.4 of Johnston (2021), A is invertible if and only if a finite number of elementary row operations transforms A into I_n or, equivalently, there exist elementary matrices $E_1, \dots, E_k$ such that

$$E_k \dots E_1 A = I_n.$$

- If A is invertible, it follows that

$$E_k \dots E_1 = E_k \dots E_1 I_n = E_k \dots E_1 A A^{-1} = I_n A^{-1} = A^{-1}.$$

- In simple terms, the elementary row operations that bring A to I_n simultaneously bring I_n to A^{-1} .

Computing the inverse of a matrix

Theorem (Theorem 2.2.5 of Johnston (2021))

Suppose $A \in \mathbb{R}^{n \times n}$. Then A is invertible if and only if there exists a matrix $E \in \mathbb{R}^{n \times n}$ such that the reduced REF of the block matrix $[A \mid I_n]$ is $[I_n \mid E]$. Furthermore, if A is invertible then it is necessarily the case that $A^{-1} = E$.

Exercise

- Suppose that

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 5 & 3 \\ 1 & 0 & 8 \end{bmatrix}.$$

- Find the inverse of A by bringing the augmented matrix

$$[A \mid I_3] = \left[\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 2 & 5 & 3 & 0 & 1 & 0 \\ 1 & 0 & 8 & 0 & 0 & 1 \end{array} \right]$$

to a reduced REF.

- If the reduced REF is of the form $[I_3 \mid B]$, then A is invertible and $B = A^{-1}$. If not, then A is not invertible.

A simplifying theorem

Theorem (Theorem 2.2.7 of Johnston (2021))

Suppose that $A \in \mathbb{R}^{n \times n}$. If there exists a matrix $B \in \mathbb{R}^{n \times n}$ such that $AB = I_n$ or $BA = I_n$, then A is invertible and $A^{-1} = B$.

Proof.

If $BA = I_n$, then multiplying the linear system $Ax = 0$ on the left by B shows that $BAX = B0 = 0$, but also $BAX = I_n x = x$, so $x = 0$. The linear system $Ax = 0$ thus has $x = 0$ as its unique solution, so it follows from Theorem 2.2.4 of Johnston (2021) that A is invertible. Multiplying the equation $BA = I_n$ on the right by A^{-1} shows that $B = A^{-1}$, as desired. If $AB = I_n$, the argument is similar (hint: consider the transpose of AB). The proof is complete. □

Matrices and matrix operations (cont.)

Systems of linear equations revisited

Unique solution of a system of linear equations

Theorem (Part (c) of Theorem 2.2.4 Johnston (2021))

If $A \in \mathbb{R}^{n \times n}$ is an invertible matrix, then for every $b \in \mathbb{R}^{n \times 1}$, the system of linear equations $Ax = b$ has exactly one solution $x = A^{-1}b$.

Proof.

Since $A(A^{-1}b) = b$, it follows that $x = A^{-1}b$ is a solution. Suppose that x_0 is another solution, i.e., $Ax_0 = b$. But then we have that $A^{-1}Ax_0 = A^{-1}b$ and $x_0 = A^{-1}b$. Hence, $A^{-1}b$ is a unique solution. The proof is complete. $\square$

Solutions of systems of linear equations

Theorem (Theorem 2.1.1 of Johnston (2021))

Every system of linear equations has either

- (a) *no solutions;*
- (b) *exactly one solution;*
- (c) *infinitely many solutions.*

Proof.

Assume that x_1 and x_2 are two solutions (i.e., $Ax_1 = Ax_2 = b$) such that $x_1 \neq x_2$ and consider another vector $(1 - c)x_1 + cx_2$ for $c \in \mathbb{R}$. Then

$$A((1 - c)x_1 + cx_2) = (1 - c)Ax_1 + cAx_2 = (1 - c)b + cb = b$$

which means the system has an infinite number of solutions since $(1 - c)x_1 + cx_2$ is a solution for any $c \in \mathbb{R}$. The proof is complete. $\square$

Matrices and matrix operations (cont.)

Diagonal and triangular matrices

Diagonal matrices

Definition

$A \in \mathbb{R}^{n \times n}$ is said to be *diagonal* if $a_{ij} = 0$ for $i \neq j$.

Examples

$$\begin{bmatrix} 2 & 0 \\ 0 & -5 \end{bmatrix}, \quad \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad \begin{bmatrix} 6 & 0 & 0 & 0 \\ 0 & -4 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 8 \end{bmatrix}, \quad \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}.$$

If $A \in \mathbb{R}^{n \times n}$ is diagonal, we also write

$$A = \text{diag}(a_1 \ \dots \ a_n).$$

Diagonal matrices (cont.)

Matrix multiplication of diagonal matrices is performed elementwise

$$\begin{aligned}\text{diag}(a_1 \ \dots \ a_n) \text{diag}(b_1 \ \dots \ b_n) &= \\ &= \text{diag}(a_1 b_1 \ \dots \ a_n b_n).\end{aligned}$$

Theorem

A diagonal matrix $D \in \mathbb{R}^{n \times n}$ is invertible if and only if $d_i \neq 0$ for all $i = 1, \dots, n$ and in which case $D^{-1} = \text{diag}(d_1^{-1} \ \dots \ d_n^{-1})$.

It is straightforward to calculate powers of a diagonal matrix D since

$$D^k = \text{diag}(d_1^k \ \dots \ d_n^k)$$

for $k \geq 1$.

Triangular matrices

Definition

$A \in \mathbb{R}^{n \times n}$ is called *upper (lower) triangular* matrix if all entries below (above) the main diagonal are zero, i.e., $a_{ij} = 0$ for $i > j$ ($a_{ij} = 0$ for $i < j$).

Examples

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ 0 & a_{22} & a_{23} & a_{24} \\ 0 & 0 & a_{33} & a_{34} \\ 0 & 0 & 0 & a_{44} \end{bmatrix} \quad \text{and} \quad \begin{bmatrix} a_{11} & 0 & 0 & 0 \\ a_{21} & a_{22} & 0 & 0 \\ a_{31} & a_{32} & a_{33} & 0 \\ a_{41} & a_{42} & a_{43} & a_{44} \end{bmatrix}.$$

Upper triangular Lower triangular

So diagonal means upper *and* lower triangular.

Properties of triangular matrices

Theorem

- (a) *The transpose of a lower triangular matrix is upper triangular.*
- (b) *The product of upper triangular matrices is upper triangular.*
- (c) *An upper triangular matrix is invertible if and only if its diagonal entries are non-zero. If so, its inverse is also upper triangular.*
- (d) *The same is true if “lower” and “upper” are swapped.*

Matrices and matrix operations (cont.)

Determinant

- Recall that for

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

we defined its determinant as $\det(A) = ad - bc$.

- We saw that A is invertible if and only if $\det(A) \neq 0$.
- We now define the determinant of any $A \in \mathbb{R}^{n \times n}$.

Inductive definition

- For $A = [a] \in \mathbb{R}^{1 \times 1}$, we define $\det([a]) := a$.
- For

$$A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix},$$

we thus have

$$\begin{aligned} \det(A) &= a_{11} \det([a_{22}]) - a_{12} \det([a_{21}]) \\ &= a_{11}a_{22} - a_{12}a_{21}. \end{aligned}$$

- General determinants are defined inductively.
- So assume that we know what the determinant of an $n \times n$ -matrix is.
- To move up in size, we need *minors* and *cofactors*.

Minors, cofactors, and determinant

Definition

If $A \in \mathbb{R}^{n \times n}$, then the (i, j) -th minor, M_{ij} , of A is the determinant of the $(n - 1) \times (n - 1)$ -matrix obtained by deleting row i and column j . The (i, j) -th cofactor is the number $C_{ij} := (-1)^{i+j} M_{ij}$.

Definition

The determinant of $A \in \mathbb{R}^{n \times n}$ is defined as

$$\det(A) = a_{11}C_{11} + a_{12}C_{12} + \dots + a_{1n}C_{1n}.$$

We also use the following notation

$$\begin{vmatrix} a_{11} & \dots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{n1} & \dots & a_{nn} \end{vmatrix} = \det \left(\begin{bmatrix} a_{11} & \dots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{n1} & \dots & a_{nn} \end{bmatrix} \right).$$

Examples

- We have that

$$\det\left(\begin{bmatrix} a & b \\ c & d \end{bmatrix}\right) = a(-1)^{1+1}d + b(-1)^{1+2}c = ad - bc.$$

- Suppose that

$$A = \begin{bmatrix} 3 & 1 & 0 \\ -2 & -4 & 3 \\ 5 & 4 & -2 \end{bmatrix}.$$

Show that $\det(A) = -1$.

Theorem (Theorem 3.2.8 of Johnston (2021))

Suppose $A \in \mathbb{R}^{n \times n}$. Then

$$\det(A) = a_{i1}C_{i1} + a_{i2}C_{i2} + \dots + a_{in}C_{in} \quad (2)$$

$$\det(A) = a_{1j}C_{1j} + a_{2j}C_{2j} + \dots + a_{nj}C_{nj} \quad (3)$$

for all $i, j = 1, \dots, n$.

- (2) is called a cofactor expansion along the i -th row.
- (3) is called a cofactor expansion along the j -th column.

Matrices and matrix operations (cont.)

Computing determinants

- In general, it is tedious to calculate the determinant of a matrix.
- When we considered systems of linear equations, we saw that certain systems of linear equations are easier to solve.
- Our goal was to transform every system of linear equations into this particular type so that we can solve it in an easier way.
- Can we do something like this with determinants as well?

Determinants of triangular matrices

Theorem (Theorem 3.2.4 of Johnston (2021))

If $A \in \mathbb{R}^{n \times n}$ is a triangular matrix, then

$$\det(A) = \prod_{i=1}^n a_{ii} = a_{11} \cdot \dots \cdot a_{nn}.$$

Example

$$\begin{vmatrix} 1 & 2 & -1 & 7 & 91 \\ 0 & 1 & 19 & 0 & \frac{1}{10} \\ 0 & 0 & 2 & 1 & -5 \\ 0 & 0 & 0 & 1 & \pi \\ 0 & 0 & 0 & 0 & 3 \end{vmatrix} = 1 \cdot 1 \cdot 2 \cdot 1 \cdot 3 = 6.$$

Recall that a diagonal matrix is also triangular.

Matrix operations and determinants

- How do various matrix operations affect the determinant?
- $\det(A) = \det(A^T)$ since column expansion of A^T is the same as row expansion in A .

Theorem (Theorem 2.2.3 of Anton, Rorres, and Kaul (2019))

Let $A \in \mathbb{R}^{n \times n}$. Then if B is the matrix obtained from A by

- (a) multiplying a row with $k \in \mathbb{R}$, then $\det(B) = k \det(A)$;*
- (b) interchanging two rows, then $\det(B) = -\det(A)$;*
- (c) adding a multiple of one row to another, then $\det(B) = \det(A)$.*

Clarification

Part (a) of Theorem 2.2.3 of Anton, Rorres, and Kaul (2019) means that

$$\det(B) = \begin{vmatrix} ka_{11} & ka_{12} & ka_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} = k \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} = k \det(A).$$

Computing the determinant

- So, if we can bring A into triangular form in a few row operations then the determinant is easy to compute, as the product down the main diagonal (see Theorem 3.2.4 of Johnston (2021)).
- But we must keep track of how it changes along the way.

Example on the blackboard

Bring

$$A = \begin{bmatrix} 0 & 1 & 5 \\ 3 & -6 & 9 \\ 2 & 6 & 1 \end{bmatrix}$$

into REF to compute the determinant of A.

Computing the determinant (cont.)

- The point is that cofactor expansion (i.e., the definition) is generally very slow and including some row operations may therefore be preferable for large matrices.
- According to Anton, Rorres, and Kaul (2019), cofactor expansion of a general 25×25 -matrix will take millions of years on a computer.
- One can mix and match the methods as one sees fit.

Tips and tricks

- Instead of row operations, one can do column operations. This corresponds to row operations on A^T .
- If a matrix A has a row or column with a lot of zeros, then this is often good for row/column cofactor expansion.
- If a matrix A has a zero row (or column) then $\det(A) = 0$.
- If two rows (columns) are proportional, then $\det(A) = 0$, since one can then create a zero-row (zero-column) by row operations.

Matrix algebra and determinants

- $\det(A + B)$ is not necessarily equal to $\det(A) + \det(B)$.
- Consider

$$A = \begin{bmatrix} 1 & 2 \\ 2 & 5 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix}.$$

Then $\det(A + B) \neq \det(A) + \det(B)$.

Theorem

If $A, B \in \mathbb{R}^{n \times n}$, then $\det(AB) = \det(A) \det(B)$.

Matrices and matrix operations (cont.)

What can determinants be used for?

Invertibility and the determinant

Theorem

A matrix $A \in \mathbb{R}^{n \times n}$ is invertible if and only if $\det(A) \neq 0$.

Proof.

If A is invertible then,

$$1 = \det(I_n) = \det(AA^{-1}) = \det(A) \det(A^{-1})$$

and hence $\det(A) \neq 0$.

Suppose that $\det(A) \neq 0$ and consider the reduced REF B of A . Row operations change the determinant, but cannot make it zero. So $\det(B) \neq 0$. This means that B cannot have a zero-row. The only $n \times n$ reduced REF with no zero rows is the identity matrix so $B = I_n$. So I_n is the reduced REF of A , so A is invertible by Theorem 2.2.4 of Johnston (2021).

Theorem (Theorem 3.A.2 of Johnston (2021))

Suppose $A \in \mathbb{R}^{n \times n}$ is invertible and $b \in \mathbb{R}^{n \times 1}$. Define A_j to be the matrix that equals A , except its j -th column is replaced by b . Then the linear system $Ax = b$ has a unique solution x , whose entries are

$$x_j = \frac{\det(A_j)}{\det(A)} \quad \text{for all } 1 \leq j \leq n.$$

Example

Use Cramer's rule to solve

$$\begin{cases} x_1 + \quad + 2x_3 = 6; \\ -3x_1 + 4x_2 + 6x_3 = 30; \\ -x_1 - 2x_2 + 3x_3 = 8. \end{cases}$$

References



Johnston, Nathaniel (2021). *Introduction to linear and matrix algebra*. Springer Cham.



Anton, Howard, Chris Rorres, and Anton Kaul (Sept. 2019). *Elementary linear algebra: applications version*. 12th edition. John Wiley & Sons, Inc.